NUMERICAL EVIDENCE FOR CONJECTURES ON NEGATIVE DISCRETE SECOND MOMENTS OF DIRICHLET L-FUNCTIONS

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Abstract. Pearce-Crump recently conjectured that, for a primitive Dirichlet character χ modulo q, $\sum_{0<\gamma\le T}|L'(\rho,\chi)|^{-2}\sim\frac{A_q}{2\pi}T$ with $A_q=\frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1}$, and $\sum_{0<\gamma\le T}|L(2\rho,\chi^2)/L'(\rho,\chi)|^2\sim\frac{\varphi(q)}{q}\frac{T}{2\pi}$. These conjectures come from a long-mollifier heuristic and generalise conjectures of Gonek and Ng for the Riemann zeta-function. We test them numerically for the real primitive characters of conductor 4, 5, 8 and 12, using between 2000 and 3000 zeros of each L-function, computed and verified for completeness, together with the first 27000 zeros of $\zeta(s)$. To remove the common finite-height bias, we compare each L-function with $\zeta(s)$ at the same local density of zeros. In all eight tests the predicted arithmetic factor lies inside the 95% bootstrap interval and the value without arithmetic factor is excluded. For conductor 12, where the two conjectures predict 1/2 and 1/3, we measure $0.501\,[0.445,0.555]$ and $0.340\,[0.301,0.381]$. All predictions and analysis choices were fixed before the corresponding computations.

1. Introduction

Let $\rho=\frac{1}{2}+i\gamma$ denote the non-trivial zeros of $\zeta(s)$. Assuming the Riemann Hypothesis and that all zeros are simple, Gonek [4] conjectured

$$\sum_{0<\gamma\le T}\frac{1}{|\zeta'(\rho)|^2}\sim\frac{3}{\pi^3}\,T, \tag{1}$$

and Milinovich and Ng [7] proved a lower bound of half this size. Ng [8] conjectured the companion asymptotic $\sum_{0<\gamma\le T}|\zeta(2\rho)/\zeta'(\rho)|^2\sim T/2\pi$, with a matching lower bound of half the size due to Sinha [10]. Negative discrete moments of this kind enter explicit formulae for the summatory functions of the Möbius and Liouville functions, and their sizes were modelled with random matrix theory by Hughes, Keating and O'Connell [6]. Numerically, Hiary and Odlyzko [5] found agreement with (1) using large sets of zeros near the 10^{16}th, 10^{20}th and 10^{23}rd zeros.

For a primitive Dirichlet character χ modulo $q>1$, Pearce-Crump [9] proved lower bounds for

$$J_\chi(T)=\sum_{0<\gamma\le T}\frac{1}{|L'(\rho,\chi)|^2},\qquad K_\chi(T)=\sum_{0<\gamma\le T}\left|\frac{L(2\rho,\chi^2)}{L'(\rho,\chi)}\right|^2,$$

uniformly in the conductor, and formulated the following conjectures [9, Conjectures 7 and 8]:

$$J_\chi(T)\sim\frac{A_q}{2\pi}\,T,\qquad A_q=\frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1}, \tag{2}$$

$$K_\chi(T)\sim\frac{\varphi(q)}{q}\,\frac{T}{2\pi}. \tag{3}$$

Both reduce to the conjectures of Gonek and Ng when $q=1$. They were obtained from Farmer's long-mollifier heuristic [3], which, as Pearce-Crump points out, is not a proof and is applied there to discrete sums rather than to the continuous mean values for which it was designed. Their derivation from the ratios conjecture [1, 2] is left open in [9].

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The content of (2) and (3) beyond the case $q = 1$ lies in the arithmetic factors $\prod_{p|q} p/(p+1)$ and $\varphi(q)/q$. The aim of this note is to test these factors numerically. We are not aware of previous numerical tests of (2) or (3); our literature search is described in Section 7.

2. Method

2.1. **Normalisation.** Since the zeros of $L(s,\chi)$ at height t have density $\frac{1}{2\pi}\log\frac{qt}{2\pi}$, (2) and (3) say, to first order, that the zero-wise averages of

$$v_n = \frac{\log(q\gamma_n/2\pi)}{|L'(\rho_n,\chi)|^2}, \qquad w_n = \frac{\log(q\gamma_n/2\pi)\,|L(2\rho_n,\chi^2)|^2}{|L'(\rho_n,\chi)|^2}$$

tend to A_q and $\varphi(q)/q$ respectively (to $6/\pi^2$ and 1 for ζ).

2.2. **A differential statistic.** At the heights accessible to us the convergence is slow: for $\zeta(s)$ the average of v_n over the zeros 1–3000, 3001–10000 and 10001–20000 equals 0.912, 0.963 and 0.992 times $6/\pi^2$. We therefore do not compare absolute averages with the conjectured constants. Instead, we compare each L-function with $\zeta(s)$ at the same local density $D = \log(q\gamma/2\pi)$. We split the zeros of $L(s,\chi)$ into ten bins at the deciles of D and define

$$R = \frac{\operatorname{mean}_\chi(v)}{\sum_b \frac{N_b}{N}\,\operatorname{mean}_{\zeta,\,b}(v)},$$

where N_b is the number of zeros of $L(s,\chi)$ in bin b and $\operatorname{mean}_{\zeta,b}(v)$ is the average of v over the zeros of ζ whose density $\log(\gamma/2\pi)$ lies in the same bin; R_8 is defined in the same way with w. If the finite-height corrections depend on the height mainly through the density, they cancel in R to first order, and (2), (3) predict

$$R = \prod_{p|q}\frac{p}{p+1}, \qquad R_8 = \frac{\varphi(q)}{q}.$$

This cancellation is an assumption, not a theorem; it is supported by the fact that the absolute ratios $\operatorname{mean}(v)/A_q$ for χ (0.932 for $q = 4$, 0.937 for $q = 5$) lie in the same range as for ζ at comparable heights.

2.3. **Alternatives.** For each test we also record two alternatives: H_0, no arithmetic factor ($R = 1$); and, for (2), the factor $\prod_{p|q} p^2/(p^2-1)$ obtained by simply removing the Euler factors at $p \mid q$ from $1/\zeta(2)$; for (3), the factor of (2), which tests whether the two moments require different factors.

2.4. **Uncertainty.** The terms $1/|L'(\rho,\chi)|^2$ have a heavy tail produced by close pairs of zeros: the largest 1% of the terms carry 12–23% of the sums. We use 95% percentile intervals from a block bootstrap (blocks of 100 consecutive zeros, resampled independently for $L(s,\chi)$ and $\zeta(s)$, 2000 replicates). Because the bootstrap may under-cover with heavy tails, we also recompute R after removing the five largest terms of $L(s,\chi)$.

2.5. **Pre-registration.** For each group of tests, the predictions, alternatives, statistics, bootstrap settings and decision rules were written down and fixed (with SHA-256 hashes) before the derivatives were computed. A test is counted as favourable if the predicted value lies in the interval and H_0 does not.

3. Data

3.1. **Zeros of $\zeta(s)$.** We use the first 27000 zeros ($\gamma \le 23476.2$) from Odlyzko's table `zeros1` (declared accuracy $3\cdot10^{-9}$). The tests for $q = 4, 5$ use the first 20000 zeros, and those for $q = 8, 12$ use 27000, so that the densities of the reference cover those of the L-functions.

3.2. **Zeros of $L(s,\chi)$.** We use the real primitive characters $\chi_{-4} = (\frac{-4}{\cdot})$ (odd), $\chi_5 = (\frac{\cdot}{5})$, $\chi_8 = (\frac{8}{\cdot})$ and $\chi_{12} = (\frac{12}{\cdot})$ (even). Zeros were located by sign changes of the real function $Z_\chi(t) = e^{i\theta_\chi(t)}L(\frac{1}{2}+it,\chi)$, with L evaluated by Hurwitz zeta sums in `mpmath` at 30 digits, a scan step of at most $1/8$ of the mean spacing, and refinement to 10^{-10}. Each list was checked for (V1) agreement of the count with $\theta_\chi(T)/\pi$ at the end points, (V2) agreement with an independent rescan at half the step, (V3) a second evaluation at random zeros, and (V4) reality of Z_χ.

Sign-change scans lose pairs of very close zeros. This happened: the original lists for χ_{-4} and χ_5 failed V1/V2 (deficits of 14 and 6 zeros in 12000), and the list for χ_8 failed V2 (2000 found against 2002 in the rescan). In every case the missing zeros were recovered by rescanning all gaps larger than 1.6 mean spacings with a step of $1/32$ of the mean spacing; they were always close pairs (for χ_8, the pair 1470.2269, 1470.2791). The corrected lists pass all checks, and the corrected χ_8 list has exactly the 2002 zeros found by the rescan. We use the zeros 1–3000 for $q = 4, 5$ ($\gamma \le 2891.3$ and 2808.1) and 1–2000 for $q = 8, 12$ ($\gamma \le 1856.6$ and 1764.1).

3.3. **Derivatives and $L(2\rho,\chi^2)$.** Since $|Z'_\chi(\gamma)| = |L'(\rho,\chi)|$ at a zero, we computed $|\zeta'(\rho)|$ with `mpmath.siegelz` and $|L'(\rho,\chi)|$ by a central difference of Z_χ with step 10^{-6} at 20 digits (relative error below $4\cdot 10^{-10}$ against step 10^{-5} at 30 digits). For real χ, χ^2 is the principal character, so $L(2\rho,\chi^2) = \zeta(1+2i\gamma)\prod_{p\mid q}(1-p^{-1-2i\gamma})$.

4. Results

q	moment	R measured	95% interval	predicted	alternatives excluded
4	J	0.655	$[0.592, 0.715]$	2/3	1, 4/3
5	J	0.827	$[0.753, 0.901]$	5/6	1, 25/24
8	J	0.687	$[0.605, 0.784]$	2/3	1, 4/3
12	J	0.501	$[0.445, 0.555]$	1/2	1, 3/2
4	K	0.490	$[0.430, 0.549]$	1/2	1, 2/3
5	K	0.753	$[0.673, 0.835]$	4/5	1 (not 5/6)
8	K	0.558	$[0.450, 0.684]$	1/2	1 (not 2/3)
12	K	0.340	$[0.301, 0.381]$	1/3	1, 1/2

Table 1. Differential statistics R (for J_χ) and R_8 (for K_χ). Removing the five largest terms gives 0.633 ($q = 4$) and 0.794 ($q = 5$) for J, without changing any conclusion.

In all eight tests the predicted factor lies inside the interval and H_0 is excluded (Table 1). In six of them the other alternative is excluded as well. In particular, for $q = 4$ and $q = 12$ the two moments require their own factors: exchanging the factors of (2) and (3) is excluded. The characters χ_8 and χ_{-4} have the same predicted factors but opposite parity, and their measured values are compatible.

For $\zeta(s)$, the average of w_n (Ng's moment, conjectured limit 1) equals 0.927, 0.948 and 0.934 on the zeros 1–3000, 3001–10000 and 10001–20000, again showing finite-height deviations of a few per cent that the differential statistic is designed to cancel.

5. A heuristic check of the arithmetic factors

The constants in (2) and (3) can also be recovered from the random matrix model of Hughes, Keating and O'Connell [6], independently of the mollifier heuristic. For $\zeta(s)$ that model predicts $\sum_{0<\gamma\le T}|\zeta'(\rho)|^{2k} \sim \frac{T}{2\pi}\frac{G^2(k+2)}{G(2k+3)}a(k)(\log T)^{k(k+2)+1}$, where G is Barnes' function and $a(k) = \prod_p(1-$

$\frac{1}{p})^{k^2}\,\mathbb{E}_\theta|1-p^{-1/2}e^{i\theta}|^{-2k}$; for $k=-1$ this gives (1). For a primitive character χ modulo q we keep the same random matrix factor and replace the local average by $\mathbb{E}_\theta|1-\chi(p)p^{-1/2}e^{i\theta}|^{-2k}$, which equals 1 when $p\mid q$, while the factor $(1-\frac{1}{p})^{k^2}$ is kept for every prime. For $k=1$ this reproduces the factor $\varphi(q)/q$ in the mean square $\int_0^T|L(\frac{1}{2}+it,\chi)|^2\,dt$, as follows from the approximate functional equation and the mean value theorem. For $k=-1$ it gives

$$\prod_{p\nmid q}\Big(1-\frac{1}{p^2}\Big)\prod_{p\mid q}\Big(1-\frac{1}{p}\Big)=\frac{6}{\pi^2}\prod_{p\mid q}\frac{p}{p+1}=A_q.$$

In the same way, the local average of $|L_p(2\rho,\chi^2)|^2|L_p(\rho,\chi)|^{-2}$ equals $1/(1-\frac{1}{p})$ for $p\nmid q$ and 1 for $p\mid q$, which gives the factor $\varphi(q)/q$ of (3). Thus two independent heuristics lead to the same constants. The alternatives excluded by the data also have a simple meaning in this picture: the factor $\prod_{p\mid q}p^2/(p^2-1)$ corresponds to dropping $(1-\frac{1}{p})$ at $p\mid q$, and exchanging the factors of (2) and (3) corresponds to using the local average of one moment for the other. This is a heuristic consistency check and says nothing about lower-order terms; we do not claim it is new.

6. Discussion and limitations

The data support the arithmetic factors in (2) and (3) for four real characters, including a composite conductor. The evidence is limited in several ways: the heights are low ($\gamma<3000$); only real characters of small conductor were used; the reference set of ζ-zeros is shared between tests; the bootstrap intervals may be optimistic because of heavy tails; and the differential statistic relies on the assumption that finite-height corrections depend on the height mainly through the density of zeros.

A derivation of (2) and (3) from the ratios conjecture, as suggested in [9], would also predict the lower-order terms. At the low heights used here those terms are of a few per cent and measurable, so the present data could test such a prediction in absolute rather than differential form. Complex characters and larger heights are natural further tests.

7. Literature search

We searched zbMATH Open, the arXiv, and the works citing Gonek's paper on negative moments (via OpenAlex) for numerical computations of $L'(\rho,\chi)$ at zeros of Dirichlet L-functions or of the sums J_χ, K_χ, and found none; the only numerical study of the corresponding moments of ζ that we found is [5]. MathSciNet was not available to us.

Data and code

The zeros, derivatives, pre-registration documents and analysis code are available from the author on request. All computations were run on a single laptop with two cores; the computation and verification of the zeros of the L-functions dominated the cost.

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